Python-Based Visualization and Machine Learning Prediction of Airline Flight Delays

Dataset: https://www.kaggle.com/datasets/whenamancodes/flight-delay-prediction

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1. Introduction

Aviation industry is an important segment of the international transport system, however, flight delays are a perennial issue that derails schedules, upsurges operational expenses and affects passenger satisfaction. The increased pressure on air flight and an increase in the complexity of the operating conditions have contributed to delay prediction being one of the tasks of a high priority. The proposed project will utilize Python-based program that incorporates advanced visualization and machine learning to improve the way of predicting flight delays and interpreting the results (Valkama, 2025). The solution proposed capitalizes on the open-source tools to give the solution flexibility, transparency, and value as a tool of analysis to all decision-makers and researchers.

1.1 Problem Definition

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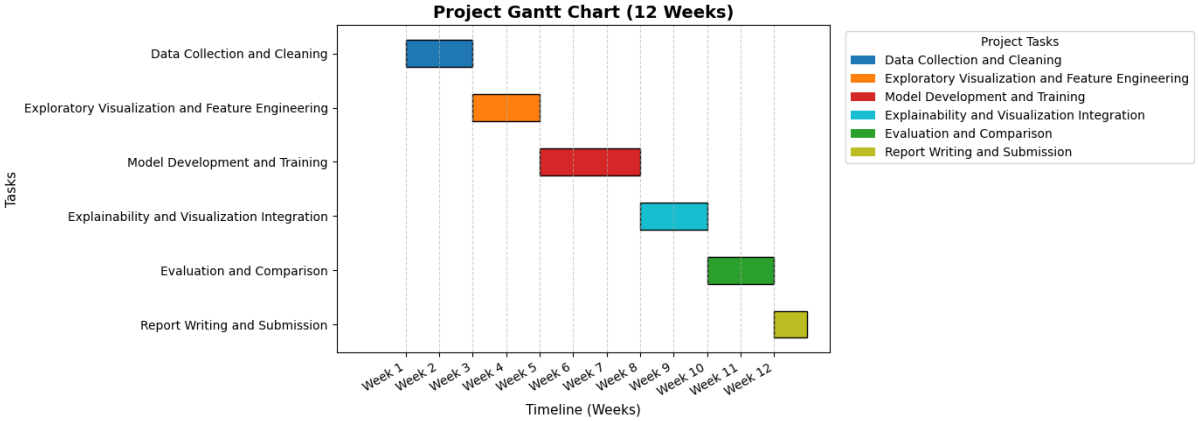

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